

ROHAN RAMPERSAD

Full-Stack Developer | AI & Automation | Database-Driven Systems

Trinidad & Tobago - Remote-Friendly | Applied Computer Science

PROJECT FOCUS

Full-stack apps + AI automation

DATABASES

MySQL + PostgreSQL + Prisma

DEPLOYED WORK

Live Vercel demo + GitHub repos

TECH DEPTH

Computer vision + ML workflows

Contact & Project Links

Portfolio Website:	https://rohanrampersad383-beep.github.io/rohanrampersad.github.io/
Main GitHub:	https://github.com/rohanrampersad383-beep
LinkedIn:	https://linkedin.com/in/rohan-rampersad-a18ab1401
AI Job Match Live Demo:	https://ai-job-match-assistant.vercel.app
AI Job Match Repo:	https://github.com/rohanrampersad383-beep/ai-job-match-assistant
Fitness Central Repo:	https://github.com/rohanrampersad383-beep/fitness-central

Applied Computer Science student and junior full-stack developer with practical experience building AI-assisted systems, automation workflows, dashboard applications, and database-driven platforms. Experienced with Next.js, TypeScript, PostgreSQL, Prisma ORM, PHP, MySQL, Python, MATLAB, Tailwind CSS, and AI/ML workflows across web development, image processing, and data-focused systems.

Frontend

HTML, CSS, JavaScript, TypeScript, Tailwind CSS, Next.js, responsive UI design

Backend & Databases

PHP, MySQL, PostgreSQL, Prisma ORM, SQL, CRUD, schema design, normalization, auth

AI / Data

Python, Jupyter Notebook, preprocessing, feature extraction, model evaluation, confusion matrices

Computer Vision

MATLAB, RGB/HSV thresholding, segmentation, morphology, GLI, centroid/medoid analysis

Cloud / Tools

Git, GitHub, Vercel, VS Code, WAMP Server, phpMyAdmin, AWS foundations, Excel

System Design

Waterfall, OOAD, ERDs, DFDs, use cases, sequence diagrams, testing, documentation

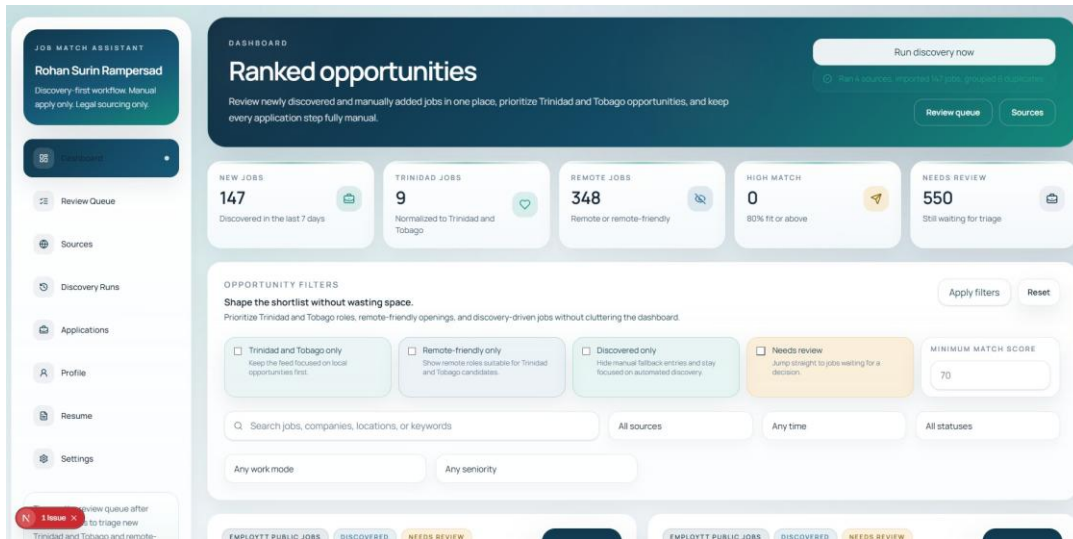
What This Portfolio Demonstrates

- Ability to build and deploy a modern full-stack application with authentication, dashboards, and database-backed workflows.
- Experience with AI/ML coursework across tabular, image, audio, and text data with model evaluation outputs.
- Computer vision and MATLAB image-processing workflow for segmentation, measurement, and ranking tasks.
- Documentation mindset: coursework reports, diagrams, project evaluations, test plans, GitHub repositories, and live deployment evidence.

Featured Project 1: AI Job Match Assistant

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Live full-stack application with job discovery, ranking, dashboards, and tracking



Dashboard - ranked opportunities and filtering workflow

[Live Demo](#)

[GitHub Repo](#)

[Portfolio Website](#)

Project Overview

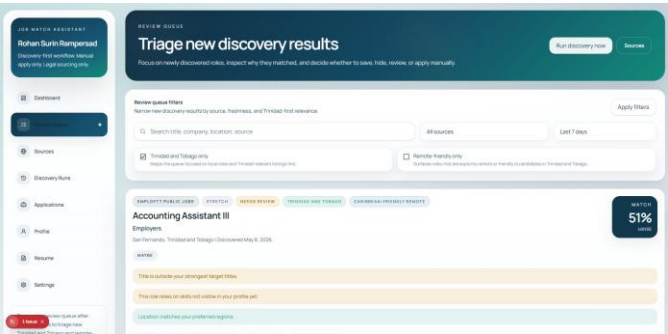
AI Job Match Assistant is a full-stack job discovery and tracking platform designed to help users discover, evaluate, organize, and track job opportunities through a legal semi-automated workflow. It focuses on Trinidad & Tobago job discovery while also supporting remote-friendly roles.

Core Technologies

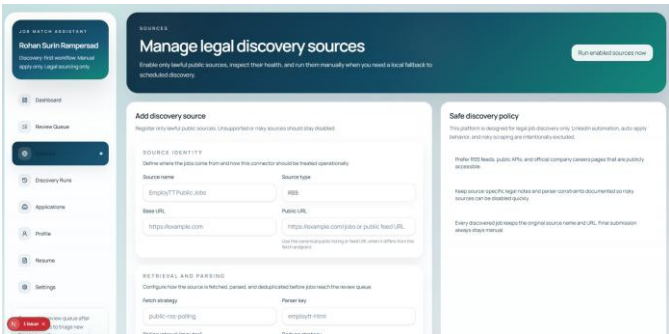
Next.js, TypeScript, Tailwind CSS, Prisma ORM, PostgreSQL, Vercel, RSS feeds, public APIs, custom scoring logic, dashboard UI components.

Key Value

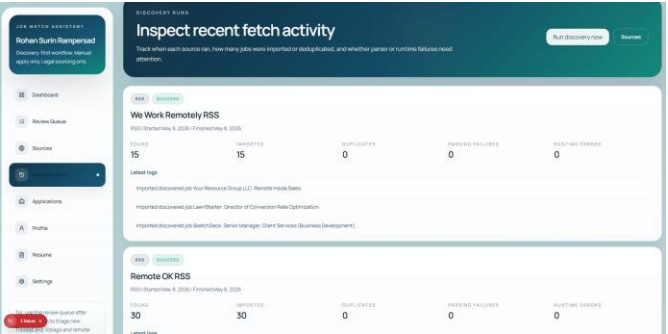
Shows modern SaaS-style development, backend data modelling, ethical automation, responsive UI design, and deployment readiness.



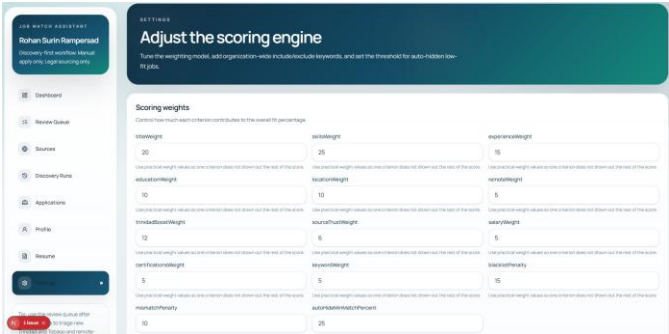
Review Queue



Sources



Discovery Runs



Scoring Engine

Discovery Dashboard

Ranks and filters opportunities by source, region, work mode, and match score.

Source Management

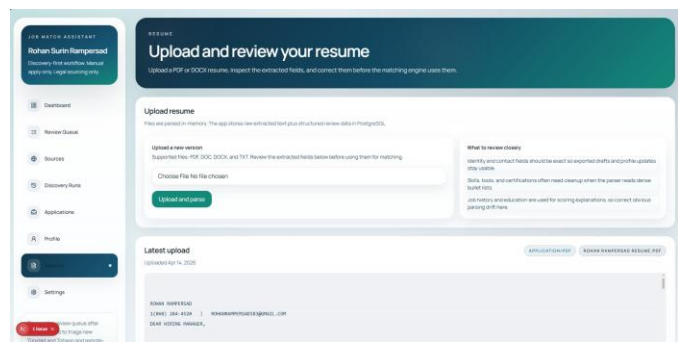
Stores lawful public sources with operational notes and safe discovery constraints.

Review Queue

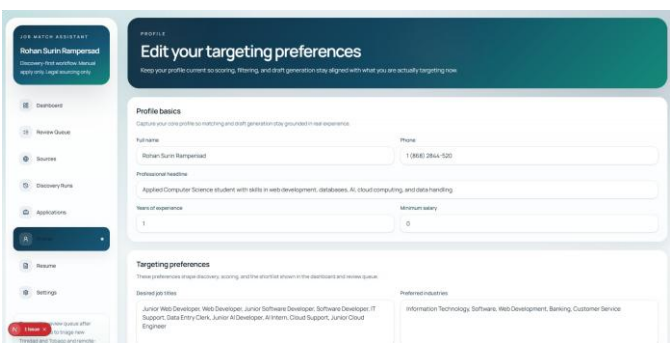
Supports manual triage so every application decision remains user-controlled.

Scoring Engine

Allows matching weights and thresholds to be adjusted for better fit evaluation.



Resume upload and extracted review



Targeting preferences and profile setup

Problem

Remote and local job discovery can become scattered across multiple job boards, RSS feeds, public listings, and manual tracking tools. Applicants need a cleaner workflow to review, score, and organize opportunities without unsafe automation.

Solution

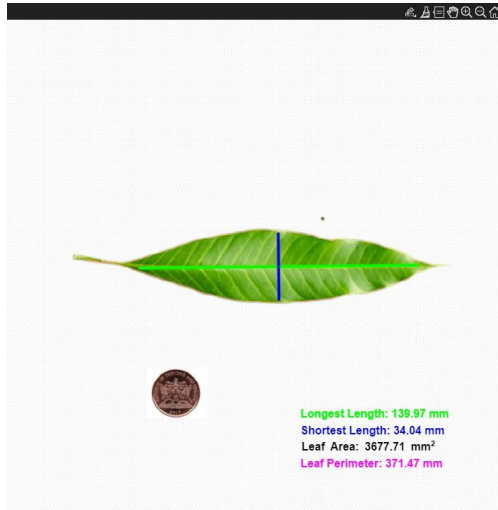
Built a full-stack platform that imports roles from approved public sources, normalizes job metadata, groups duplicate listings, scores fit, and keeps final application steps manual.

Architecture Highlights

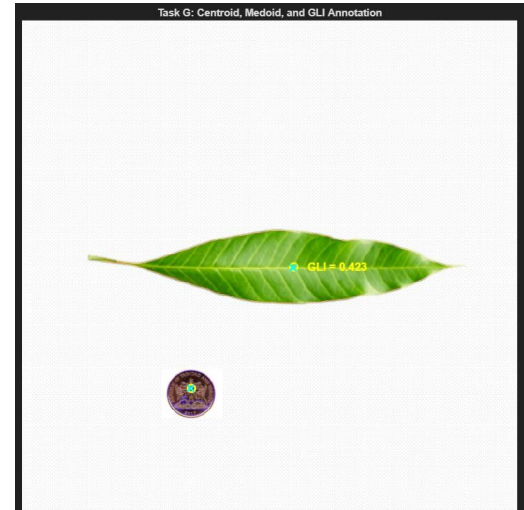
Next.js application structure, PostgreSQL relational data models, Prisma ORM, source adapter concepts, dashboard filtering, scoring settings, resume parsing support, and Vercel deployment.

Professional Relevance

Demonstrates full-stack development, data-driven workflows, ethical automation, UI/UX design, backend architecture, and remote-ready project documentation.



Morphometric measurements with real-world scaling



Centroid, medoid, and GLI annotation

Project Overview

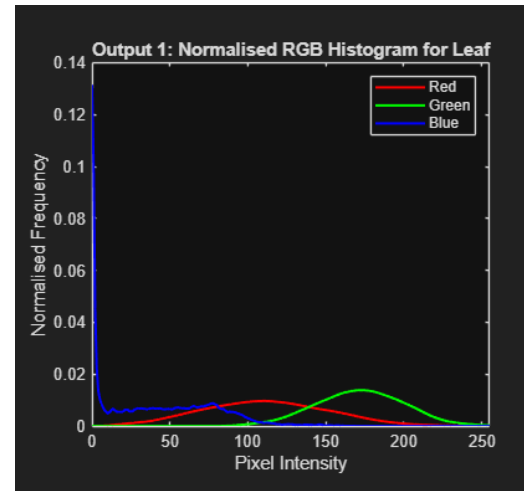
This MATLAB image-processing project used computer vision techniques to process leaf and coin images, isolate objects, extract biological measurements, and compare leaves using quantitative features. The workflow included RGB/HSV thresholding, binary masks, morphological filtering, boundary detection, centroid and medoid calculations, Green Leaf Index analysis, histograms, and multi-leaf rankings.

Techniques Demonstrated

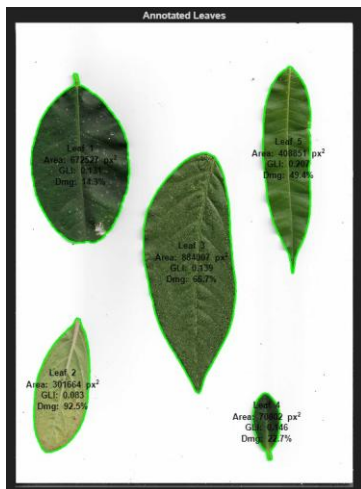
- RGB/HSV thresholding and object segmentation
- Morphological cleaning with hole filling and object filtering
- RGB histogram analysis and color feature extraction
- Centroid, medoid, GLI, area, perimeter, and axis measurements
- Multi-leaf ranking by greenness, area, and damage



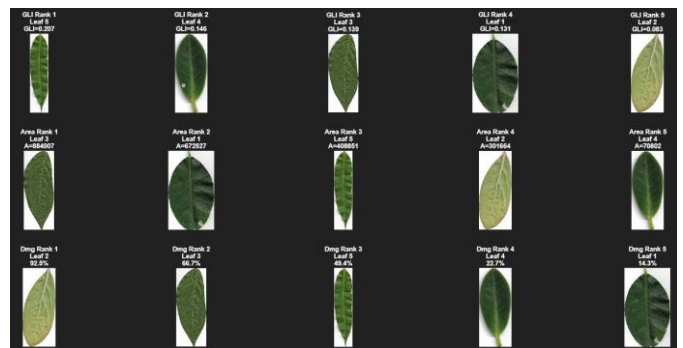
Segmentation Output



RGB Histograms



Annotated Leaves



Multi-Leaf Ranking

Why This Project Matters

This project adds technical depth beyond standard web development. It shows ability to work with visual data, transform raw images into measurable features, and build analysis workflows that produce interpretable outputs. It strengthens the portfolio for roles involving AI, data, image processing, automation, and analytical programming.

Certification Support

MATLAB Onramp completion supports the MATLAB and image processing skills used in this project.

Project Overview

Completed practical AI/ML notebooks across tabular, image, audio, and text data. Work included preprocessing, feature extraction, model training, classification, confusion matrices, evaluation metrics, and visual interpretation of results.

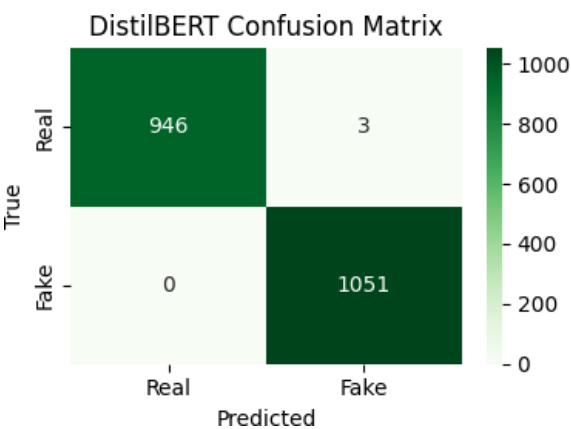
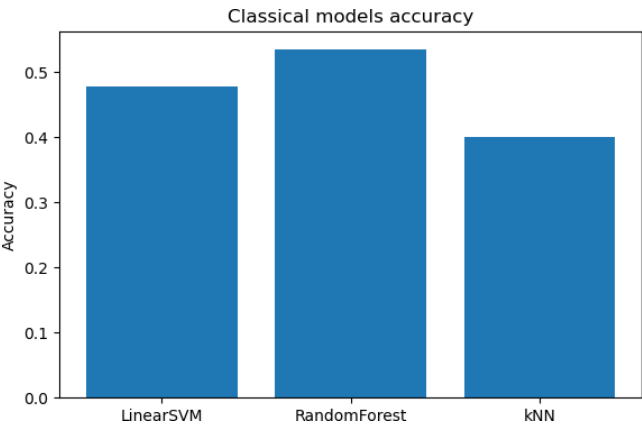
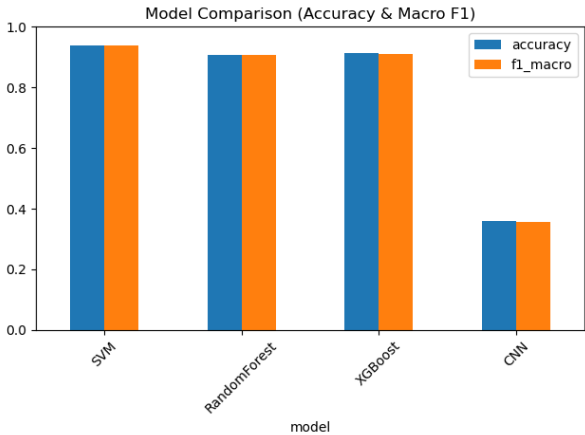
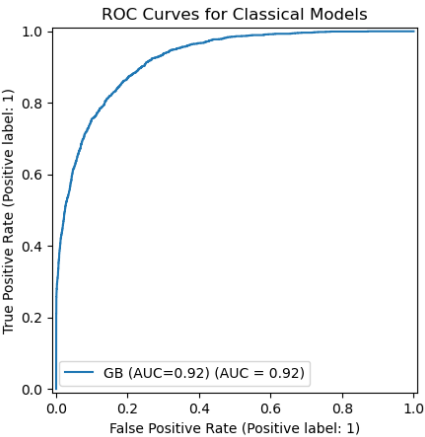


Image Model Accuracy

NLP Confusion Matrix



Tabular ROC Curves

Audio Model Comparison

Fitness Central - Web-Based Fitness Tracking System

HTML, CSS, JavaScript, PHP, MySQL, WAMP Server

Built a browser-based fitness tracking system with registration/login, workout logging, weight records, goal setting, dashboard progress, and an exercise guide. Supporting documentation included Waterfall SDLC, OOAD diagrams, database design, testing, and evaluation. Repository: <https://github.com/rohanrampersad383-beep/fitness-central>

Database Design Projects - Velocity Motors and Toyota

MySQL, SQL, ERDs, DFDs, Data Dictionaries

Completed database design and system analysis projects covering ER modelling, normalization, SQL scripts, data dictionaries, query design, business rules, reports, interface planning, and evaluation.

BIT Project - Mandy's Hardware Inventory / Client Management

Systems Analysis, DFDs, ERDs, CRUD Planning

Designed inventory and client management documentation including requirements, risk management, cost-benefit analysis, DFD, ERD, SQL table planning, interface design, query design, and walkthrough evaluation.

Computer Networks / IT Infrastructure

Subnetting, VPNs, Cisco IOS, Windows, Troubleshooting

Covered topology planning, UDP traffic, server roles, Wi-Fi concepts, site surveys, PoE access points, IPv4 subnetting, security, VPNs, scalability, IoT risks, Windows installation, antivirus setup, and system maintenance.

Cloud Computing and AWS Practical Preparation

Practical exposure to AWS IAM, VPC networking, EC2, EBS, RDS security, load balancers, Auto Scaling, security groups, route tables, user data scripts, CLI interpretation, and troubleshooting configuration issues.

Business, Ethics, and Professional Studies

Coursework included reports on deepfakes, privacy, misinformation, e-business models, ERP/CRM/SCM systems, operations management, value chains, leadership, financial analysis, and business problem solving.

Accounts Clerk - Persads Supercenter

Jun 2024 - Jan 2025

- Managed daily financial records and data entry using Microsoft Excel.
- Tracked sales, payroll, daily income reports, and cash flow balancing.
- Maintained accuracy through transaction checks and structured reporting.
- Built transferable strengths in data accuracy, confidentiality, organization, and problem solving.

Cashier / Inventory Assistant - Mandy Sooklal Hardware

Jun 2022 - Sep 2022

Processed customer transactions accurately while supporting stock updates, restocking, and basic inventory monitoring. Developed communication, customer service, teamwork, and operational organization skills in a busy retail environment.

Education

- BSc (Hons) Applied Computer Science - Anglia Ruskin University, delivered through SAMS - Expected Apr 2026
- NCC Education Level 5 Diploma in Computing / Computer Business - SAMS - Completed 2025
- NCC Education Level 4 Diploma in Computing / Computer Business - SAMS - Completed 2024
- High School Diploma - Cowen Hamilton Secondary School - 2017-2022

Certifications

- MATLAB Onramp Certificate - MathWorks Training Services - Rise Virtual Employment Certificate - July 2025



MATLAB Onramp Certificate

Target Remote Roles

- Junior Full-Stack Developer
- Junior Web Developer
- AI Automation Assistant
- IT Support Technician
- Data Operations Assistant
- Database Support Trainee

Career Direction

My short-term goal is to gain hands-on experience in software development, IT support, data operations, web development, databases, cloud computing, or technical support. My long-term goal is to grow into a stronger technology professional capable of building larger systems, improving business workflows, and contributing to practical software and automation solutions.

Recruiter Keywords

Next.js, TypeScript, PHP, MySQL, PostgreSQL, Prisma, Python, SQL, REST APIs, GitHub, authentication, dashboards, CRUD, Excel reporting, MATLAB, computer vision, AI/ML.

Rohan Rampersad - Technical Portfolio Summary

Applied Computer Science student and junior full-stack developer with practical experience in full-stack web development, AI/ML coursework, MATLAB image processing, database design, systems analysis, cloud foundations, and data-focused work.

Main Strengths

- Deployed full-stack job discovery app using Next.js, TypeScript, Prisma, PostgreSQL, and Vercel.
- Computer vision project using MATLAB segmentation, morphology, GLI, histograms, measurements, and rankings.
- AI/ML notebook work across tabular, image, sound, and text data with evaluation visuals.
- Database and systems analysis background with ERDs, DFDs, SQL, normalization, documentation, and testing.

Key Project Links

Live Demo:	https://ai-job-match-assistant.vercel.app
AI Job Match Repository:	https://github.com/rohanrampersad383-beep/ai-job-match-assistant
Main GitHub:	https://github.com/rohanrampersad383-beep
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Fitness Central Repository:	https://github.com/rohanrampersad383-beep/fitness-central

Final Positioning

Best suited for remote-friendly junior roles that value practical development, fast learning, clear documentation, data accuracy, and willingness to build and improve real systems.

[View Live AI Job Match Demo](#)[View GitHub](#)